

Study Organizer (NOVA/Jarvis) Project Context & App Structure

This document provides a comprehensive overview of the study_organizer (also referred to as NOVA or Jarvis) codebase. You can use this as a system prompt or structural guide to rapidly understand the scope, features, and architecture of this Flutter project so you can easily modify and extend it.

1. Project High-Level Overview

App Description: A futuristic, highly-interactive study management application built with Flutter. It encompasses AI features (like a conversational assistant named Jarvis/NOVA), productivity tools (a "Detroit 2060" themed Pomodoro timer called "Cognitive Reactor"), subject/topic organization, schedule generation, text-to-speech feedback, voice activity detection (VAD), and background tracking to ensure focus.

Core Themes & Features:

Study & Task Management: Subjects, Topics, Tasks, Semesters, Exams, Marks, and Reminders.

AI & Intelligence (Jarvis/NOVA):

ElevenLabs integration for high-quality Text-to-Speech (

NovaElevenLabsService.dart

).

Groq and Gemini integrated for intelligence, schedule building, and conversational capabilities (

nova_intelligence_engine.dart

,

jarvis_brain_service.dart

,

ai_service.dart

).

Voice Activity Detection (VAD) using speech_to_text for a continuous wake word (

NovaVadService.dart

).

Interactive UI (Detroit 2060 / Futuristic):

Custom overlays (

jarvis_overlay.dart

,

contamination_overlay.dart

,

knowledge_xray_overlay.dart

,

lightning_drill_overlay.dart

).

Cognitive Reactor (Pomodoro system) redesigned with a "Detroit 2060" futuristic, deep-focus aesthetic (

cognitive_reactor_page.dart

).

PDF text extraction, markdown rendering with LaTeX support (flutter_markdown_latex, flutter_math_fork).

Background & Watchdog Processes:

workmanager and app_usage for background tasks and distraction monitoring (

nova_watchdog_service.dart

).

geolocator for location-based brief triggers (

nova_location_service.dart

).

2. Tech Stack & Dependencies (

pubspec.yaml

)

State Management: flutter_bloc & equatable

Database: sqflite

AI / LLM: google_generative_ai (Gemini), http (Groq API calls)

Audio & TTS: flutter_tts, audioplayers, record (recording Audio), ElevenLabs APIs

UX / UI: table_calendar, fl_chart, confetti, flutter_markdown, syncfusion_flutter_pdf

Background / Hardware: workmanager, app_usage, geolocator, speech_to_text, shared_preferences

3. Directory Structure (lib/)

Based on the latest file tree, the application follows a robust modular structure:

lib/models/ (Data Entities)

Models represent the core data schemas parsed usually directly from SQL or JSON APIs.

Core: subject.dart, topic.dart, task.dart, semester.dart, mark.dart, subject_note.dart, subject_metadata.dart.

Scheduling/Focus: timetable.dart, pomodoro_session.dart, reminder.dart, exams.dart.

AI: jarvis_document.dart.

lib/bloc/ (State Management)

Contains the global application state.

app_bloc.dart, app_state.dart, app_event.dart. Note: Check how events dispatch state changes across the complex UI overlays.

lib/services/ (Core Business & AI Logic)

This is the heavily AI-driven backend layer.

AI & Brain:

jarvis_brain_service.dart / jarvis_service.dart: The core conversational and orchestration logic.

nova_intelligence_engine.dart: Advanced analysis and reasoning.

Nova_smart_schedule_service.dart: AI-generated study scheduling.

ai_service.dart: Base AI communication.

Voice & Audio:

NovaElevenLabsService.dart: Handles high-fidelity TTS streaming.

NovaVadService.dart: Voice Activity Detection / wake word listener.

nova_audio_service.dart: General sound effects and TTS fallback.

Context & App Integration:

jarvis_navigator.dart: AI-driven app navigation logic.

nova_location_service.dart: Home geofencing for daily briefs.

nova_brief_service.dart: Daily AI summaries.

nova_watchdog_service.dart: Distraction detection in the background.

Database: database.dart (sqlite initialization and CRUD wrappers).

Notifications: notifications.dart.

lib/pages/ (Main Screens)

main_shell.dart: Top-level navigational shell.

dashboard.dart: The primary hub.

subjects.dart, subject_detail.dart, topics.dart, tasks.dart, marks.dart: CRUD and list views for study data.

cognitive_reactor_page.dart: The immersive Detroit 2060 Pomodoro/Focus timer.

study_plan_page.dart, daily_schedule_page.dart, weekly_briefing_page.dart: Planning and schedule visualizations.

campus.dart: Likely a social or mapping feature.

nova_settings_page.dart: API Key configuration (Groq/Gemini) and TTS configuration.

exam_prep_hud.dart: A specialized HUD style UI for exam period studying.

lib/widgets/ (Components & Overlays)

Overlays (The "HUD" elements):

jarvis_overlay.dart: Primary AI interaction interface (microphone, TTS visuals).

knowledge_xray_overlay.dart: For specific AI analysis formatting.

lightning_drill_overlay.dart, night_before_overlay.dart, contamination_overlay.dart: Specialized visual modes for different study states.

Micro-components: nova_brief_card.dart, jarvis_button.dart, jarvis_quiz_page.dart, glass.dart (glassmorphism UI wrapper), atext.dart, helpers.dart.

4. How to Approach Developing or Modifying This App

If you (the AI) are asked to modify this codebase, keep the following context in mind:

AI Interactivity: The user values the "Jarvis" metaphor. Any response, text-to-speech function, or schedule generation should feel "smart". Ensure jarvis_brain_service.dart and nova_intelligence_engine.dart are correctly interacting with NovaElevenLabsService.dart or flutter_tts.

Futuristic Aesthetics: Recently updated to a "Detroit 2060" aesthetic (especially the Cognitive Reactor timer). UI changes should respect clean, 3D-inspired, dynamic and "holographic" or sleek animations (e.g. glass.dart and the _overlay.dart files).

Background Services: The app uses workmanager and app_usage heavily. Because background execution on iOS/Android is tricky with Flutter, be cautious when diagnosing errors inside nova_watchdog_service.dart or background workers.

Recent Work/Fixes (Context):

Added Groq and Gemini API key inputs to nova_settings_page.dart.

Stabilized TTS and microphone capture (NovaVadService.dart).

Cleaned up obsolete features (removed "Gap Detector" and "Mark Surgeon").

Transformed the timer (cognitive_reactor_page.dart) to have an away-timer grace period causing phone vibrations (breach notification) if the user leaves the app when the screen is on.

Refined AI analysis by fixing ASCII arrays into proper Markdown tables and bullet points.

5. Next Steps for Analyzing Code

Whenever the user asks you to implement a feature or fix a bug in a specific area:

Read database.dart if introducing new fields.

Read app_bloc.dart if cross-page state updates are needed.

Modify /services/ for business logic (AI generation/timing rules) and /pages/ or /widgets/ for UI mapping.

Review pubspec.yaml to ensure no conflicting library versions are introduced.

here is the project structure:

```
lib/
|
|— main.dart      # App entry point (initializes app and services)
|
|— app.dart       # Main App root widget and global Theme configuration
|
|— new.dart       # Auxiliary/Draft implementation file
|
|— bloc/          # Global state management and event handling
|   |— app_bloc.dart # Main business logic controller
|   |— app_event.dart # Definitions for user & system actions
|   |— app_state.dart # Definitions for application UI states
|
|— models/        # Data Layer (Entities/PODOs)
|   |— exams.dart
|   |— jarvis_document.dart
|   |— mark.dart
|   |— pomodoro_session.dart
|   |— reminder.dart
|   |— semester.dart
|   |— subject.dart
|   |— subject_metadata.dart
|   |— subject_note.dart
|   |— task.dart
|   |— timetable.dart
|   |— topic.dart
|
```

```
└─ services/          # Core Logic, Integrations & Infrastructure
|   └─ ai_service.dart      # Base LLM communication
|   └─ database.dart        # SQLite helper & CRUD operations
|   └─ jarvis_brain_service.dart  # Core AI orchestration logic
|   └─ jarvis_navigator.dart  # AI-driven navigation logic
|   └─ jarvis_service.dart    # Voice and primary interaction management
|   └─ notifications.dart     # Local notification handling
|   └─ NovaElevenLabsService.dart  # High-quality TTS integration
|   └─ NovaVadService.dart    # Voice Activity Detection (Wake Word)
|   └─ nova_audio_service.dart  # General audio/sound playback
|   └─ nova_brief_service.dart  # Daily briefing generation
|   └─ nova_intelligence_engine.dart # Advanced reasoning and analysis
|   └─ nova_location_service.dart  # Location/geofencing for triggers
|   └─ Nova_smart_schedule_service.dart # AI schedule generation
└─ nova_watchdog_service.dart  # Distraction monitoring/background tasks

└─ pages/             # Major Views & Feature Screens
|   └─ dashboard.dart      # Main user hub
|   └─ main_shell.dart      # App navigational shell (HUD)
|   └─ subjects.dart        # Subject overview list
|   └─ subject_detail.dart   # Detailed view for a specific subject
|   └─ topics.dart          # Topics list/management
|   └─ tasks.dart           # Tasks list/management
|   └─ marks.dart           # Grades/marks tracking
|   └─ calendar.dart        # Calendar view
|   └─ daily_schedule_page.dart  # Daily agenda view
|   └─ study_plan_page.dart   # AI-generated study plan view
|   └─ weekly_briefing_page.dart  # Weekly summary view
|   └─ cognitive_reactor_page.dart # Interactive Focus/Pomodoro Timer
|   └─ exam_prep_hud.dart     # Specialized exam preparation mode
|   └─ campus.dart          # Campus/social features
└─ nova_settings_page.dart    # Settings (API keys, TTS config)

└─ widgets/           # Reusable UI Components & Overlays
|   └─ atext.dart          # Custom text widget
|   └─ helpers.dart        # UI helper functions
```

└─ glass.dart # Glassmorphism aesthetic wrapper

└─ jarvis_button.dart # Primary interaction button

└─ jarvis_quiz_page.dart # Interactive quiz component

└─ nova_brief_card.dart # Briefing summary card

└─ jarvis_overlay.dart # Primary AI HUD overlay

└─ contamination_overlay.dart # Visual state/alert overlay

└─ knowledge_xray_overlay.dart # AI analysis formatting wrapper

└─ lightning_drill_overlay.dart # Fast-paced quiz overlay

└─ night_before_overlay.dart # Crunch-time study overlay